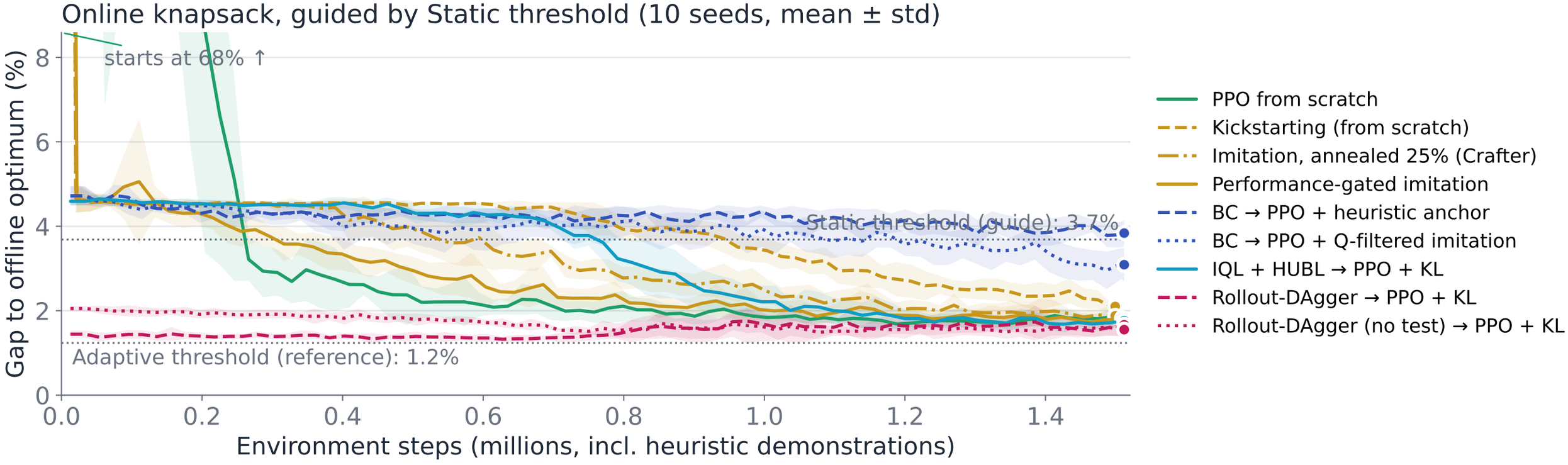
Online knapsack, guided by Static threshold (10 seeds, mean ± std)
starts at 68% ↑
Static threshold (guide): 3.7%
Adaptive threshold (reference): 1.2%
Gap to offline optimum (%)
0
2
4
6
8
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Environment steps (millions, incl. heuristic demonstrations)
PPO from scratch
Kickstarting (from scratch)
Imitation, annealed 25% (Crafter)
Performance-gated imitation
BC → PPO + heuristic anchor
BC → PPO + Q-filtered imitation
IQL + HUBL → PPO + KL
Rollout-DAgger → PPO + KL
Rollout-DAgger (no test) → PPO + KL